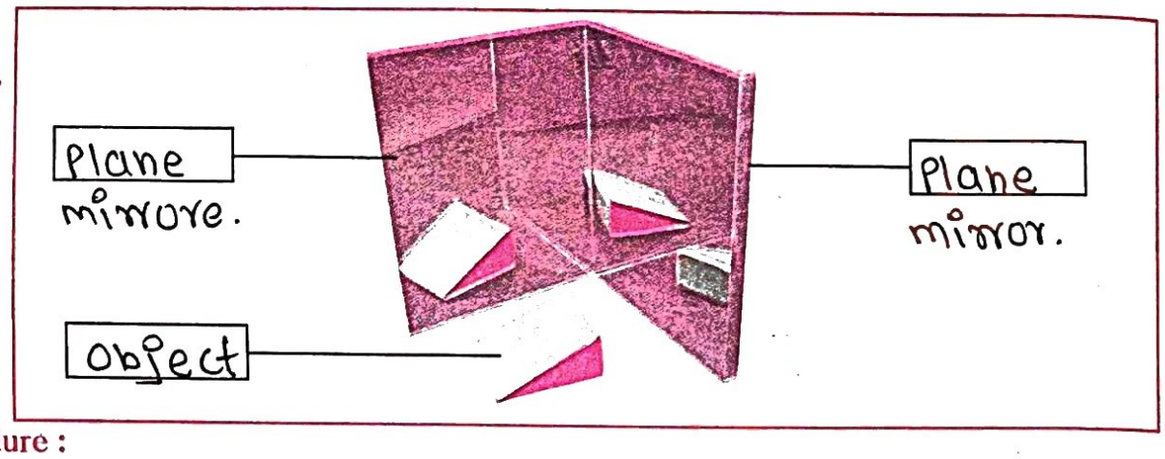
# Experiment No.8

**Title** : To study the formula of number of images from the number of images formed by changing the angle between two mirrors.

**Apparatus** : Two big plane mirrors, Stand to hold the mirrors, one object, Protractor etc.

**Figure :**




**Procedure :**

1. Put two big plane mirrors in the stand properly .
2. Adjust with the help of protractor an angle 30° between the two mirrors.
3. Place an object between two mirrors.
4. Count the total number of images formed in the two mirrors.
5. Repeat the steps 2 to 4 in the procedure by keeping the two mirrors at an angle of $45^0$, $60^0$, $90^0$ and $180^0$

**Formula :**

$$n = \frac{360}{A} - 1$$

**Observation Table :**

| Sr.No | Angle | Number of Images n | Number of Images as per the formula |
|---|---|---|---|
| 1 | $30^0$ | 11 | $n = \frac{360}{30} = 12 - 1 = 11$ |
| 2 | $45^0$ | 7 | $n = \frac{360}{45} = 8 - 1 = 7$ |
| 3 | $60^0$ | 5 | $n = \frac{360}{60} = 6 - 1 = 5$ |
| 4 | $90^0$ | 3 | $n = \frac{360}{180} = 4 - 1 = 3$ |
| 5 | $180^0$ | 1 | $n = \frac{360}{180} = 2 - 1 = 1$. |

**Inference :**

There is a ..$n = \frac{360°}{\theta} - 1$.....relationship between the number of images and the angle between the mirrors.

.....There the angles between the mirror image
In the images dicreases and when if decases.....

## Multiple Choice Questions

1. The number of images ....., when the measure of the between the two mirror is increased.

   A. decreases B. remains the same C. increases D. different answer than these

2. ..... images are obtained when the plane mirrors are kept parallel to each other.

   A. 2 B. 4 C. 1 D. infinite

3 The image of the mirror kept at 10 cm in front of plane mirror will be obtained at the distance .....

   A. 5 cm behind the mirror B. 10 cm behind the mirror

   C. 10 cm before the mirror D. 5 cm before the mirror

## Questions

1. Explain the formula verified experimentally as above regarding the images obtained in plane mirrors.

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.....Image produced will be vitual and the really in-verted and of same as a source.....

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2. If a quarter of a circle is placed between two plane mirrors, what should be the angle between the two mirrors so that of a complete circle is seen with the help of images?

.....The I mass in front of mirror which can be obtained on a screen is called image or an position of the object a conavere mirror always produce on errect demished image.....

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